Publication List – Sean Patrick MacBride

Publication list current as of October 23, 2026. h-index of 8, as of October 23, 2026.

1. Rodeghiero, G., Drlica-Wagner, A., Taranto, A., Rosignoli, L., Pollek, H., Pai, A., Jones, L., Howard, E., MacBride, S., Andrew, J., *et al.* An overview of stray light findings and interpretation during on-sky commissioning of LSSTCam. *arXiv preprint arXiv:2606.31945* (2026).
2. Abac, A., North, C., Bhandare, R., Pierra, G., Amato, A., Chen, D., Haskell, B., Robinet, F., Fyffe, M., Muccillo, L., *et al. arXiv: GWTC-5.0: Methods for Identifying and Characterizing Gravitational-wave Transients* tech. rep. (2026).
3. Wüthrich, J., Zhao, G., Yamani, B., Lebrun, L., MacBride, S., Fazan, A., Andrade-Oliveira, F. & Soares-Santos, M. Characterizing robotic positioners under the influence of changing gravity vectors for future spectroscopic surveys. *arXiv preprint arXiv:2606.26821* (2026).
4. MacBride, S. P., McMahon, I., Laghi, D., Sánchez-Cid, D., Andrade-Oliveira, F., Soares-Santos, M., Bilicki, M., Ferraiuolo, S., More, S., Gray, R., *et al.* Constraints on the Cosmic Expansion Rate Using the DECam Local Volume Exploration Survey and the Fifth Gravitational Wave Transient Catalog. *arXiv preprint arXiv:2610.25909* (2026).
5. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adhicary, S., Adhikari, D., Adhikari, R., Adkins, V., *et al.* Constraints on ultralight bosons from merging binary and remnant black holes observed during the second and third parts of the fourth LIGO-Virgo-KAGRA observing run. *arXiv preprint arXiv:2608.11620* (2026).
6. Frész, G., Simcoe, R., Angile, E., Molina, G. B., Egan, M., Gabutti, M., Lazar, J., Malonis, A., Mastersson, R., Mitchell, B., *et al.* First year of LLAMAS the large lenslet array Magellan spectrograph: design, build, commissioning, first results, and lessons learned. In *Ground-based and Airborne Instrumentation for Astronomy XI* **14149,** 141491L (2026).
7. Collaboration, L. S., Collaboration, V. & Collaboration, K. GWTC-5.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation. *arXiv preprint arXiv:2605.27227* (2026).
8. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adhicary, S., Adhikari, D., Adhikari, R., Adkins, V., *et al.* GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog. *arXiv preprint arXiv:2605.27225* (2026).
9. Scientific, L., Van den Bossche, E., *et al.* GWTC-5.0: Observations from the second part of the fourth LIGO-Virgo-KAGRA observing run and updates to the Gravitational-Wave Transient Catalog. *Astrophysical Journal Letters* **1004** (2026).
10. Collaboration, L. S., Collaboration, V., Collaboration, K., *et al.* GWTC-5.0: Observations from the second part of the fourth LIGO-Virgo-KAGRA observing run and updates to the Gravitational-Wave Transient Catalog. *arXiv e-prints,* arXiv–2605 (2026).
11. Sun, L., Kuns, K., Slagmolen, B., Fritschel, P., Schmidt, P., Lantz, B., Chua, S., Ballmer, S., Barton, M., Cumming, A., *et al.* LIGO A: Detector Design and Science Prospects Beyond A+. *arXiv preprint arXiv:2608.11673* (2026).
12. Fanning, K., Antilogus, P., Aubourg, E., Banovetz, J. D., Bernard, A., Bond, T., Boucaud, A., Bowdish, B., Bregeon, J., Breugnon, P., *et al.* LSSTCam installation and commissioning at the Vera C. Rubin Observatory. In *Ground-based and Airborne Telescopes XI* **14147,** 406–418 (2026).
13. McMahon, I., Laghi, D., Soares-Santos, M., Ackley, K., Dálya, G., Gençel, Y., Sánchez-Cid, D., Andrade-Oliveira, F., MacBride, S., Chapman-Bird, C., *et al.* Measurement of the Hubble constant using the Dark Energy Survey Year 6 Gold galaxy catalog and the fourth Gravitational-Wave Transient Catalog. *arXiv preprint arXiv:2602.04766* (2026).

14. Malagón, A. A. P., Ordenes-Briceño, Y., Taranu, D. S., Broughton, A., Choi, Y., Carlin, J. L., Reed, S., Ramírez, K. P., MacBride, S. P., Tohfa, H. M., *et al.* Multi-Method Machine Learning Classification of Astronomical Sources with Rubin Data Previews: a Spectroscopy-Free Approach with DP1 and DP2. *arXiv preprint arXiv:2610.20746* (2026).

15. Chandler, C. O., Bernardinelli, P. H., Juri, M., Singh, D., Hsieh, H. H., Sullivan, I., Jones, R. L., Kurlander, J. A., Vavilov, D., Eggl, S., *et al.* Nsf-doe vera c. rubin observatory observations of interstellar comet 3i/atlas (c/2025 n1). *The Astrophysical Journal Letters* **1001,** L35 (2026).

16. collaboration, L. S., collaboration, V., collaboration, K., *et al.* Open Data from LIGO, Virgo, and KAGRA through the Second Part of the Fourth Observing Run. *arXiv e-prints,* arXiv–2605 (2026).

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18. Lebrun, L., Wüthrich, J., Fazan, A., MacBride, S., Andrade-Oliveira, F. & Soares-Santos, M. Precision measurements of the axial drift in fiber positioners for spectroscopic surveys using defocus blur. In *Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation VII* **14154,** 1785–1796 (2026).

19. Jones, R. L., Yoachim, P., Neilsen Jr, E. H., Ribeiro, T., Dennihy, E. & MacBride, S. P. Scheduling the LSST: commissioning the Rubin feature-based scheduler. In *Observatory Operations: Strategies, Processes, and Systems XI* **14151,** 378–394 (2026).

20. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adhicary, S., Adhikari, D., Adhikari, R., Adkins, V., *et al.* Search for Compact Binary Coalescences with Sub-solar Mass Components in Data from the Second Part of LIGO-Virgo-KAGRA's Fourth Observing Run. *arXiv preprint arXiv:2610.20276* (2026).

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22. Developers, R. O. S. P., AlSayyad, Y., Armstrong, R., Axelrod, T. S., Banovetz, J., Bechtol, K., Bellm, E. C., Bernardinelli, P. H., Bosch, J. F., Bradshaw, A., *et al.* The LSST Science Pipelines Software: Optical Survey Pipeline Reduction and Analysis Environment. *arXiv preprint arXiv:2610.22613* (2026).

23. MacBride, S. P., Roodman, A., Marshall, S., Utsumi, Y., Fanning, K., Banovetz, J., Schutt, T., Broughton, A., Liang, S., Rasmussen, A. P., *et al.* The On-Sky Performance of the LSST Camera CCD Array. *arXiv preprint arXiv:2606.31806* (2026).

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26. Juramy, C., Antilogus, P., Astier, P., Banovetz, J., MacBride, S. P., Rasmussen, A. P. & Utsumi, Y. The" dark dips" phenomenon in the LSST Camera on-sky images. *arXiv preprint arXiv:2607.00925* (2026).

27. Abac, A., Abe, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adam, A., Adamcewicz, C., Adhicary, S., Adhikari, D., Adhikari, R., *et al.* Updated Upper Limits on the Isotropic Gravitational-Wave Background from LIGO, Virgo, and KAGRA Data through April 2025. *arXiv preprint arXiv:2608.23477* (2026).

28. Zhang, H., Kokubo, M., MacBride, S., McMahon, I., Tominaga, N., Utsumi, Y., Yoshida, M., Morokuma, T., Tanaka, M., Arai, A., *et al.* A Joint Search for the Electromagnetic Counterpart to the Gravitational-Wave Binary Black-Hole Merger Candidate S250328ae with the Dark Energy Camera and the Prime Focus Spectrograph. *arXiv preprint arXiv:2508.00291* (2025).

29. Abac, A., Abouelfettouh, I., Acernese, F., Ackley, K., Adamcewicz, C., Adhicary, S., Adhikari, D., Adhikari, N., Adhikari, R., Adkins, V., *et al.* GWTC-4.0: Population properties of merging compact binaries. *arXiv preprint arXiv:2508.18083* (2025).

30. Team, L. C., Antilogus, P., Astier, P., Banovetz, J., Bregeon, J., Chiang, J., Digel, S. W., Fanning, K., Faris, Y., Guillemin, T., *et al.* LSST Camera Electro-Optical Test (Run 7) Results. *Commissioning Technical Note SITCOMTN-148, NSF-DOE Vera C. Rubin Observatory (August 2025)* (2025).

31. MacBride, S. P. RTN-098: Target-of-Opportunity Operations During the Science Verification Surveys. *NSF-DOE Vera C. Rubin Observatory Technical Report,* 93 (2025).

32. Team, L. C., Antilogus, P., Astier, P., Banovetz, J., Bregeon, J., Chiang, J., Digel, S. W., Fanning, K., Faris, Y., Guillemin, T., *et al.* SITCOMTN-148: LSST Camera Electro-Optical Test (Run 7) Results. *Rubin Observatory Technical Report,* 76 (2025).

33. Chandler, C., Kurlander, J., Bernardinelli, P., Juri, M., Eggl, S., Singh, D., Schwamb, M. E., Hsieh, H., Sullivan, I., Jones, R. L., *et al.* Solar System Science with the NSF-DOE Vera C. Rubin Observatory: Overview and Results. In *AGU Fall Meeting Abstracts* **2025,** P32B–05 (2025).

34. Borghi, N., Moresco, M., Anderson, R. I., Carbone, C., Cimatti, A., Escoffier, S., Giocoli, C., MacBride, S., Majidi, F. Z., Milakovi, D., *et al.* Standard Sirens in 2040s: Probing the Cosmic Expansion History with Gravitational Waves and Spectroscopic Galaxy Surveys. *arXiv preprint arXiv:2512.18369* (2025).

35. Besuner, R., Dey, A., Drlica-Wagner, A., Ebina, H., Moroni, G. F., Ferraro, S., Forero-Romero, J., Honscheid, K., Jelinsky, P., Lang, D., *et al.* The Spectroscopic Stage-5 Experiment. *arXiv preprint arXiv:2503.07923* (2025).

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37. Simcoe, R. A., Furesz, G., Masterson, R., Stenzel, J., Frostig, D., Piortowski, J., Egan, M., Malonis, A., Hinrichsen, E., Gabutti, M., *et al.* Construction update on the Magellan LLAMAS integral field spectrograph. In *Ground-based and Airborne Instrumentation for Astronomy IX* **12184,** 121840D (2022).